

Duplex

A two-channel compressor, de-esser and peak limiter for stereo, dual-mono or mid-side work.



Overview

Duplex is a dual-channel dynamics processor in the spirit of the classic 80s British two-channel units. Each channel is a clean VCA compressor with a gentle 6 dB soft knee, a frequency-conscious de-esser and a separate fast peak limiter.

The two channels can work three ways. In STEREO, channel A runs both sides together. In DUAL, A handles the left side and B the right. In M-S, A handles the mid (the centre of the image) and B the side (the edges), so you can compress the middle and the width separately.

The detector measures the average level over a few milliseconds rather than single peaks. AUTO timing follows the music: the attack speeds up the further the signal goes over, and the release lets go quickly after short peaks and slowly after long passages of compression.

Reach for it for

- Vocals: compression and de-essing in one place
- Gentle glue on a stereo bus
- Taming sibilance with SPLIT de-essing
- Compressing the centre and the sides of a mix separately (M-S)

- Dual-mono work on two unrelated sources
- Parallel drum compression

Controls

CHANNEL A and CHANNEL B

The two channels have the same controls. The label beside each name shows its job in the current MODE: BOTH SIDES, LEFT or RIGHT, MID or SIDE. In STEREO mode channel B rests and its panel says so.

IN	Switches the channel in or out. With it out, the audio passes straight through that channel.
THRESH	The level where compression starts, -40 to 0 dB (-20 dB by default).
RATIO	How strongly the channel compresses, 1:1 to INF:1 (3:1 by default). Fully right is INF:1 - limiting.
ATTACK	0.1 to 50 ms (5 ms by default). Reads AUTO when AUTO is on.
RELEASE	20 ms to 2 s (200 ms by default). Reads AUTO when AUTO is on.
GAIN	Make-up gain, -10 to +20 dB.
DE-ESS (knob)	The de-ess frequency, 2 to 12 kHz (6 kHz by default).
LIMIT	A separate fast peak limiter after the compressor, -24 to 0 dB: instant attack, 50 ms release. Fully right (0 dB) is OFF. The red light beside it shows it working.
AUTO	Program-dependent timing, replacing ATTACK and RELEASE: the further over the threshold, the faster the attack; the release is quick after short peaks and slows down after long passages of compression.
DE-ESS (button)	The compressor listens only above the DE-ESS frequency, so it reacts to sibilance rather than to the whole voice.

SPLIT	On by default; used with DE-ESS. The gain reduction is applied only above the de-ess band (the split sits at three-quarters of the DE-ESS frequency, so the whole ess band is caught): the esses come down and the body of the voice is untouched. Off: the whole signal is turned down, as on the original units.
GR	The row of 12 lights under each channel shows its gain reduction (compressor and limiter together), from 1 to 30 dB.

MASTER

MODE	STEREO: channel A runs both sides, with one shared detector. DUAL: A = left, B = right. M-S: A = mid, B = side.
LINK	Always lit in STEREO. In DUAL and M-S it is a switch (on by default): the two detectors share the louder level, so the image stays put, and the two channels' controls are ganged - turning one moves the other by the same amount, and any offset you set while unlinked is kept.
MIX	Parallel compression, 0 to 100% (shown as WET): blends the dry signal back in. 100% is fully processed.
IN / OUT meters	Input and output level. Click them to reset the peak hold.

The header

PRESETS	Opens the list of factory presets (or click the preset display). The ◀ / ▶ keys step through them. Choosing one sets both channels and the MASTER section. The last entry resets everything to the defaults.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Duplex has no input or output fold-out. An input guard is on by default, gently easing very hot sources down towards about -18 dBFS average and catching fast peaks above -6 dBFS; it can be switched off from REAPER's parameter list (Input guard).

Presets

Gentle Stereo Bus	STEREO, 2:1 at -24 dB with AUTO timing and 3 dB of gain.
Vocal Compressor	STEREO, 3:1 at -28 dB, 5 ms attack, 180 ms release, the limiter at -3 dB.
De-esser (split)	STEREO, a fast 6:1 de-esser at 6.5 kHz with SPLIT, so only the esses come down.
Broadcast Voice	STEREO, 4:1 with wideband de-essing at 5 kHz and the limiter at -2 dB.
Dual Mono Pair	DUAL, unlinked, the same 3:1 AUTO settings on each side.
M-S Glue	M-S, unlinked: gentle 2.5:1 on the mid, firmer 4:1 on the sides to hold them in.
M-S Widen + Tame Sides	M-S, unlinked: gentle AUTO compression on the mid, a 7 kHz split de-esser on the sides with 3 dB of gain.
Brickwall Safety	STEREO, INF:1 at -12 dB with a 0.3 ms attack and the limiter at -1 dB.
Parallel Drums	STEREO, 10:1 at -36 dB with 12 dB of gain, blended in at 40% mix.
Bass Leveller	STEREO, 4:1 with AUTO timing, 4 dB of gain and the limiter at -2 dB.
Reset all to defaults	Every control back to its starting value.

Tips

- For de-essing, switch DE-ESS on, sweep the DE-ESS knob until the GR lights react mostly to the esses, then set THRESH so they come down by a few dB.
- Leave SPLIT on for vocals: only the top end is turned down, so the voice doesn't dip on every 's'.
- In M-S, a little extra GAIN on channel B (side) widens the mix; compressing B harder than A holds the edges in.
- With LINK on in DUAL or M-S, set one channel and the other follows; switch LINK off to give the two channels different settings, then switch it back on to move them together.

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